

S.No.	Problem Statement	Description
1	Design and prototype a low-cost, community-friendly early flood prediction and alert system that can warn households 1–2 hours before potential flooding.	Your challenge is to build a solution that uses simple, affordable sensors (ultrasonic, water-level, pressure, or DIY IoT devices) to continuously monitor water levels in drains, nalas, river edges, or basements. The system should detect unusual or rapidly rising water patterns and predict the possibility of flooding using either rule-based logic or lightweight machine learning models.
2	CampusFlow: Digitizing Administrative Processes for Modern Colleges.	Most colleges still depend on paper-based processes for certificates, fee receipts, attendance, approvals, and grievances, forcing students into long queues, repeated office visits, and slow manual signatures for even basic documents like bonafide or mark sheets. This outdated system also burdens administrative staff with heavy paperwork, duplicate entries, verification delays, and frequent errors due to the absence of centralized digital records. During admission or exam seasons, the manual workload multiplies, causing misinformation, lost files, and unnecessary chaos. There is a clear need for a fast, transparent, and fully digital platform that replaces paperwork, reduces waiting time, improves record accuracy, and gives students and staff a modern, efficient campus administration experience.
3	Build a secure, unified digital health platform that allows patients to store, access, and share their complete medical history across hospitals, while also providing a system to verify the authenticity of medicines before use.	In most regions, a patient’s medical history is scattered across different hospitals, clinics, labs, and private practitioners. There is no unified record that follows the patient, forcing doctors to treat without complete information and causing repeated tests, misdiagnoses, and delays in critical care. Patients often have no digital access to their own prescriptions, allergies, past treatments, or reports — especially during emergencies when this information is most needed. At the same time, many areas face a serious issue of fake, expired, or counterfeit medicines , with no easy way for patients or pharmacists to verify drug authenticity. This leads to dangerous health risks, financial loss, and a major trust gap in the healthcare system. These two problems — fragmented health records and unverifiable medicines — together highlight the need for a secure, accessible, and transparent digital ecosystem.

4	Build an intelligent subscription-management and hidden-charges detection platform that helps users track, analyze, and prevent unexpected fees across OTT platforms, apps, banking services, and digital subscriptions.	Your solution should automatically scan transactions, subscription renewals, and billing patterns to identify hidden charges, silent auto-renewals, unnecessary add-ons, and long-forgotten recurring payments. The system must alert users before renewals occur, provide clear cost breakdowns, and flag suspicious or misleading charges. Additional features like simplified cancellation, budgeting insights, or comparison of alternative plans are encouraged.
5	Design a smart, real-time system that helps authorities and parents quickly identify and locate lost children in crowded public areas.	Every year, thousands of children get separated from their families in crowded places such as festivals, malls, markets, religious gatherings, railway stations, and tourist attractions. In these moments, identifying the child quickly becomes extremely difficult due to noise, crowd movement, lack of centralized information, and slow manual announcements. Security teams often rely on verbal descriptions, which may be inaccurate or delayed, and parents struggle to navigate large spaces efficiently. These delays increase panic, risk, and vulnerability. There is an urgent need for a real-time, tech-enabled system that can assist families and authorities in locating lost children faster and more accurately.
6	Build an intelligent, unified cybersecurity platform that helps users identify phishing links, understand which apps or websites track their personal data, and detect if their personal information has been leaked or exposed online.	Digital users today face three major security challenges: they cannot identify suspicious or deceptive links, they have little awareness of which apps or websites silently track their personal data, and they have no practical way to know if their phone numbers, emails, passwords, or documents have been leaked online. As a result, people—especially students, working professionals, and elderly users—unknowingly fall for phishing attacks, install privacy-invasive apps, and continue using compromised accounts without realizing the risks. Traditional antivirus tools don't provide real-time clarity, and manually checking data breaches or privacy policies is confusing and time-consuming. There is a strong need for a user-friendly, real-time, all-in-one security system that can detect risky links, expose hidden data-tracking behaviors, and monitor leaked personal information across the web.

- 7 Build a weather-based crop advisory platform specifically designed for hill and mountain farmers, offering real-time guidance based on local climate, altitude, soil conditions, and terrain-specific risks.
- Farmers living in hilly and mountainous regions face highly unpredictable weather patterns such as sudden rainfall, temperature drops, frost, landslides, soil erosion, and rapid climate shifts. Traditional flat-land farming advice does not apply to hill terrain, where altitude, slope, micro-climates, and soil variation significantly affect crop growth. Due to lack of localized weather forecasts and expert guidance, hill farmers often make incorrect decisions about sowing, irrigation, fertilizer use, pest control, and harvest timing — leading to major crop losses and unstable income. There is a strong need for a **location-specific, real-time, weather-aware advisory system** that can help hill farmers make smart agricultural decisions based on rapidly changing conditions.
- 8 Forest Fire Early Detection & Response System using IoT and AI
- Forest fires in hilly and rural regions often go unnoticed until they spread widely, causing massive ecological and economic damage. Traditional detection methods rely on manual reporting or satellite images that identify fires too late. There is a pressing need for a **real-time, automated, and accurate forest fire detection system**.